

AYMANE AIT DADS

Simulation Acceleration Intern | Geometric Deep Learning · PyTorch · Physics-Informed ML

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

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Hiring Team - Internship: Speeding up Simulations

Damen Shipyards · Gorinchem, Netherlands

Dear Damen RD&I Hiring Team,

Your Fast Physics project targets something precise: replacing hours of **high-fidelity CFD simulation** with a learned model that predicts physical quantities directly from geometry. That is exactly the tradeoff I navigated when building a **transformer autoencoder** for unsupervised fault detection on industrial audio - designing a model that had to generalize to unseen anomaly types without labeled data, evaluated on a rigorous DCASE-style benchmark, achieving **AUC 0.80** against state-of-the-art unsupervised baselines.

Two results map directly to what this role demands. First, on **training, validation, and extension across multiple domains**: I fine-tuned **CLIP ViT-L/14 (428M parameters)** on **250K samples spanning 25+ generator types**, running systematic ablation studies across unfreezing depth, learning rate schedules, and augmentation strategies - ranking **#1 on the EURECOM private leaderboard**. Second, on pipeline ownership under real constraints: at Orange Maroc I engineered an end-to-end ML pipeline over **100K+ telemetry records**, cutting manual analysis time by **~60%** and mapping outputs directly to business decisions - the same discipline needed to turn a CFD surrogate prototype into something usable in early-stage ship design.

Damen's ambition to become the world's most digitally connected shipyard - and the RD&I team's specific focus on **geometric deep learning** for hull resistance prediction - is a rare combination of rigorous physics and production ML that I want to work inside, not just read about. I would welcome the chance to discuss how my background fits the Fast Physics project.

Sincerely,

AYMANE AIT DADS